

Anand Singh

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Education

University of Texas, Dallas

- Master of Science, Computer Engineering

August 2023 - May 2025

GPA: 3.5

Dr. A.P.J. Abdul Kalam Technical University, IN

- Bachelor of Technology in Computer Science and Information Technology

August 2019 - July 2023

CGPA: 7.56 (1st Division)

Work Experience

Graduate Researcher, University of Illinois – Urbana-Champaign, IL

April 2025 - Present

- Conducting research under **Prof. Tong Zhang** as Graduate Researcher on **diffusion models**.
- Implementing Diffusion models for Document Restoration of images for better **OCR** and document digitization.
- Experimented with Denoising Diffusion Implicit Models (**DDIM**) and Diffusion Exponential Integrator Sampler (**DEIS**) to enhance sampling efficiency in diffusion-based generative models.
- Designed a two-stage training **pipeline**: synthetic dataset pre-training followed by real-world fine-tuning.

Machine Learning Engineer, Rebecca Everlene Trust Company – Chicago, IL

August 2025 - Present

- Managed the end-to-end development of the JOG App, an AI-powered medication reminder solution.
- Leading cross-functional team (AI/ML, UX/UI, Health Advisors) to deliver an innovative health-tech product.
- Overseeing the integration of key AI features like Personalized voice reminders and sentiment analysis.
- Ensuring the app's design prioritized emotional effectiveness and minimal cognitive load for elderly users.

Software Engineer Intern, Cognizant Technology Solutions – Pune, MH

January 2023 - August 2023

- Built a secure user authentication and admin dashboard for a ride-sharing platform, leveraging **Agile/Scrum**.
- Integrated **SaaS**-based API workflows and **automated** testing with Postman.
- Collaborated on a 4-member development team, improving sprint and reducing production issues by **20%**.

Publications

- **Diffusion Document Restoration** | [arxiv](#)

Under Review

Authors: Anand Singh, Shivanshu Shekhar

Conference: CVPR 2025

About: Implemented **Diffusion Models** with custom U-Net for better image extraction and OCR readability **advanced** computer vision and **deep learning** techniques.

- **Energy-Aware Models for Image Generation** | [arxiv](#)

Under Review

Authors: Shivanshu Shekhar, Alexi Gladstone, Anand Singh, Averrrryyy, Tong Zhang

Conference: TMLR 2025

About: Implemented YOLOv4 for real-time object detection on live video streams, enhancing data **extraction accuracy** from low-resolution, grainy, and foggy images using **advanced** computer vision and **deep learning** techniques.

- **Diffusion Document Restoration** | [springer](#)

July 2024

Authors: Anand Singh, Shivam Agarwal, Abhishek Kr. Gupta, Shivangi Tyagi

Conference: ICRTC 2023

About: Implemented YOLOv4 for real-time object detection on live video streams, enhancing data **extraction accuracy** from low-resolution, grainy, and foggy images using **advanced** computer vision and **deep learning** techniques.

Achievements And Awards

- **\$5000 Research Grant** and GPU resources awarded for independent research by StrongCompute, an Australia-Based Company specializing in Multi-GPU Parallelization and Cloud-based Servers.

September 2025

- SOF International Mathematics Olympiad
 - International Rank 182
 - International Rank 72

- SOF National Science Olympiad
 - Silver Medalist, Class Rank 2
 - National Rank 63

- Black Belt - Taekwondo

Projects

Medicine Management and AI Voice Reminder Application | [GitHub](#) August 2025 - Present

- Built pill recognition system using CNNs (MobileNetV2, ResNet50) with transfer learning.
- Developed personalized voice reminders via TTS and voice cloning (Hugging Face, OpenAI APIs).
- Implemented adaptive reminder scheduling using reinforcement learning and multi-armed bandits.
- By Project end, Health insights and caregiver reports with regression, boosting models, and NLP summarization.

Document Restoration Using Diffusion | [GitHub](#) April 2025 - Present

- Trained diffusion model on Doc3D dataset using CLIP-based conditioning with DDPM and DDIM; DDIM performed better.
- Designed custom U-Net encoder-decoder to predict backward map (BM) for restoration.
- Replaced CLIP encoder with custom U-Net for conditioning, improving output quality.
- Optimized with EBLO and MSE loss to enhance backward map reconstruction accuracy.

Multi-Objective RL for Dialogue Optimization | [GitHub](#) February 2025 - May 2025

- Integrated **saliency**-based attention mechanisms into Microsoft's LLM-Rubric pipeline to enhance token-level **interpretability**.
- Reimplemented and fine-tuned a **feedforward** calibration model using GPT-3.5 and Claude 3.5 response data for subscore alignment.
- Designed and executed **logit-drop** ablation tests to evaluate explanation faithfulness across 8 rubric dimensions.
- Built evaluation pipeline using **RMSE**, **Pearson's**, and **Spearman's** correlation to ensure model reproducibility.

Generative Adversarial Network's For Fashion | [GitHub](#) December 2024 - February 2025

- Built a custom GAN architecture for fashion image synthesis, including both **convolutional** generator and **discriminator** networks.
- Designed a deep convolutional generator using layers like **Dense**, **UpSampling2D**, and with **LeakyReLU** activations and Sigmoid output.
- Implemented a custom training loop with dynamic **noise injection** and loss functions to stabilize generation.
- Integrated real-time performance tracking with callbacks for **loss monitoring** and epoch-based image outputs.

Skills

Languages: C, C++ , Python, Java, Linux, MySQL.

Libraries: PyTorch, HuggingFace, Tensorflow, Pandas, NumPy, SciPy, OS, Seaborn.

Technologies: Autoencoders, Large Language Models, Git, Amazon Web Services, Docker, Apache Spark.

Certifications: AWS Machine Learning - Specialty; AWS Cloud Solutions Architect - Associate

Coursework

- Graduate:
 - Advanced Data Structures and Algorithms
 - Machine Learning and Pattern Recognition
 - Big Data Management and Analytics
 - Natural Language Processing
 - Database Design
 - Reconfigurable Systems
- Undergraduate:
 - Machine Learning Techniques
 - Discrete Structures and Logic
 - Software Development Methods
 - Mathematics IV
 - Physics
 - Cloud Computing

Extracurriculars

- Innogeeks
 - Club Coordinator and Mentor
 - Domain: Machine Learning and Android Development
- FC KIET
 - Right Center Back - Defender
 - Trainer for Recruits